Fiscal Year:	FY 2017	**Task Last Updated:**	FY 07/13/2017
PI Name:	Cheng, Zhengdong Ph.D.		
Project Title:	Liquid Crystals of Nanoplates		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77843-0001	**Congressional District:**	17
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	09/01/2013	**End Date:**	08/31/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	Liquid Crystal Facility NOTE: Extended to 8/31/2020 per NSSC information (Ed., 2/14/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AQ60G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The central objectives of the proposed research are to investigate the dynamics of phase transitions, including the formation of liquid crystals as well as the equilibrium gel or empty liquid state, of highly anisotropic colloidal disks. For the FY2016 and FY2017, ground based research was focused on the following tasks : 1. Studying the effect of nanoplate size to thickness ratio on isotropic-nematic phase transition kinetics 2. Studying the effect of surface modification of nanoplates: achieving a first order isotropic to lamellar transition instead of a isotropic to nematic transition 3. Studying the electric field induced alignment of nanoplates 4. Studying the temperature gradient induced nucleation of nematic phase within an isotropic sample and thermophoretic motion of nematic tactoids.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The understanding of the structure and properties of colloidal disks are important to soil science and space exploration (e.g., Mars), to the fundamental investigation of discotic liquid crystals, as well as various industries, including oil recovery, the delivery of pharmaceuticals, nutrition, and cosmetics. We investigated liquid crystals of colloidal platelets are very intriguing from the fundamental science perspective. Besides, external field induced optical effects may find applications in optical devices.
Task Progress:	1. Effect of nanoplate size to thickness ratio on isotropic-nematic phase transition kinetics: Liquid crystals of nanoplates take a certain amount of time to undergo transition from homogenized meta-stable state to liquid crystal state. The kinetics of the nematic liquid crystal phase formation proceeds via nucleation of nematic droplets, called tactoids, and their growth followed by inevitable sedimentation on earth bound experiments. After sedimentation of tactoids, a clear interface between isotropic liquid and nematic phase is established. We studied how long does it take for the formation of this interface (t*) by systematically varying the aspect ratio, i.e., thickness to diameter ratio of nanoplates. It was found that the larger the aspect ratio, the faster is the I-N phase separation in gravity. The tactoid formation rate is governed by the rotational and translational diffusion rates, which are faster at larger aspect ratios. The time required for I-N separation (t*) varies inversely with the mean aspect ratio (< zeta >) of nanoplates and follows the relation, t* = alpha < zeta > n, where alpha = 0.97 ± 1.30 s and n = -2.1 ± 0.2. The phase separation kinetics studied in our experiments offer guidance for the selection of optimum aspect ratio of nanoplates to conduct experiments in limited time available at the International Space Station (ISS)--constraints such as experiments in 8 hour shift that were imposed on some of the flight experiments can be accommodated. 2. First order isotropic to lamellar transition in surface modified nanoplates: Isotropic to nematic phase transition in nanoplates is very well known. However, isotropic to lamellar transition is new and was achieved using ZrP nanoplates by grafting polymers on their surface and suspending in an organic solvent. Above a certain concentration of nanoplates, tactoids of iridescent lamellar phase are observed to nucleate and grow. The color of the lamellar phase is obviously due to Brag reflection and hence is dependent on the viewing angle; however, more interestingly, the color of the lamellar tactoids is found to be dependent on the platelet aspect ratio (t/D). 3. Electric field induced alignment of nanoplates: a. Isotropic --> Anti-nematic phase transition in presence of external in-plane electric field Very dilute suspensions of nanoplates are optically isotropic and appear dark when kept between crossed polarizers. However, when an external electric field was applied, the texture appeared bright because the nanoplate aligned with the electric field and the suspension turned birefringent; this alignment is known to be anti-nematic because the directors (nanoplate normal) aligned perpendicular to the external electric field as opposed to rod-like molecules that align parallel to the field. b. Alignment of nematic directors, defect splitting in presence of through electric field and subsequent defect annihilation -- When AC voltage of 12V is applied to the nematic liquid crystal of nanoplates kept between two ITO coated glass with separation of 120 µm, the directors align perpendicular to the field and the typical nematic cross defect becomes sharp and after a while undergoes rotation of nematic brushes which most likely is due to the imbalance of elastic constants. The four brush defects later split into two brush defects which is a sign of biaxial nematic phase and is yet to be confirmed with more tests. Upon turning of the electric field, the two brush defects undergo annihilation. 4. Temperature gradient: When two ends of a 50 mm long capillary containing an isotropic ere maintained at 328K and 278K, nematic phase nucleated at the hot end as a result of thermal diffusion of nanoplates towards the hot end. The nematic phase grew larger as more nanoplates diffused to the hot end. This implied that the nanoplates have negative thermal diffusion coefficient. The direction of thermal diffusion of ZrP-TBA nanoplates (towards hot end) is opposite to that of other colloids such PMMA hard spheres. It is possible that the counter-ions in the suspension such as Bu4N+ and OH- showed normal thermal diffusion towards the cold end and created a less osmotic pressure region at the hot zone which forced nanoplates to diffuse to the hot end. The tactoids of nematic phase also showed thermal diffusion to the hot zone. It is very interesting to see the effect of gravity and thermal gradient both on the motion of tactoids.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
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